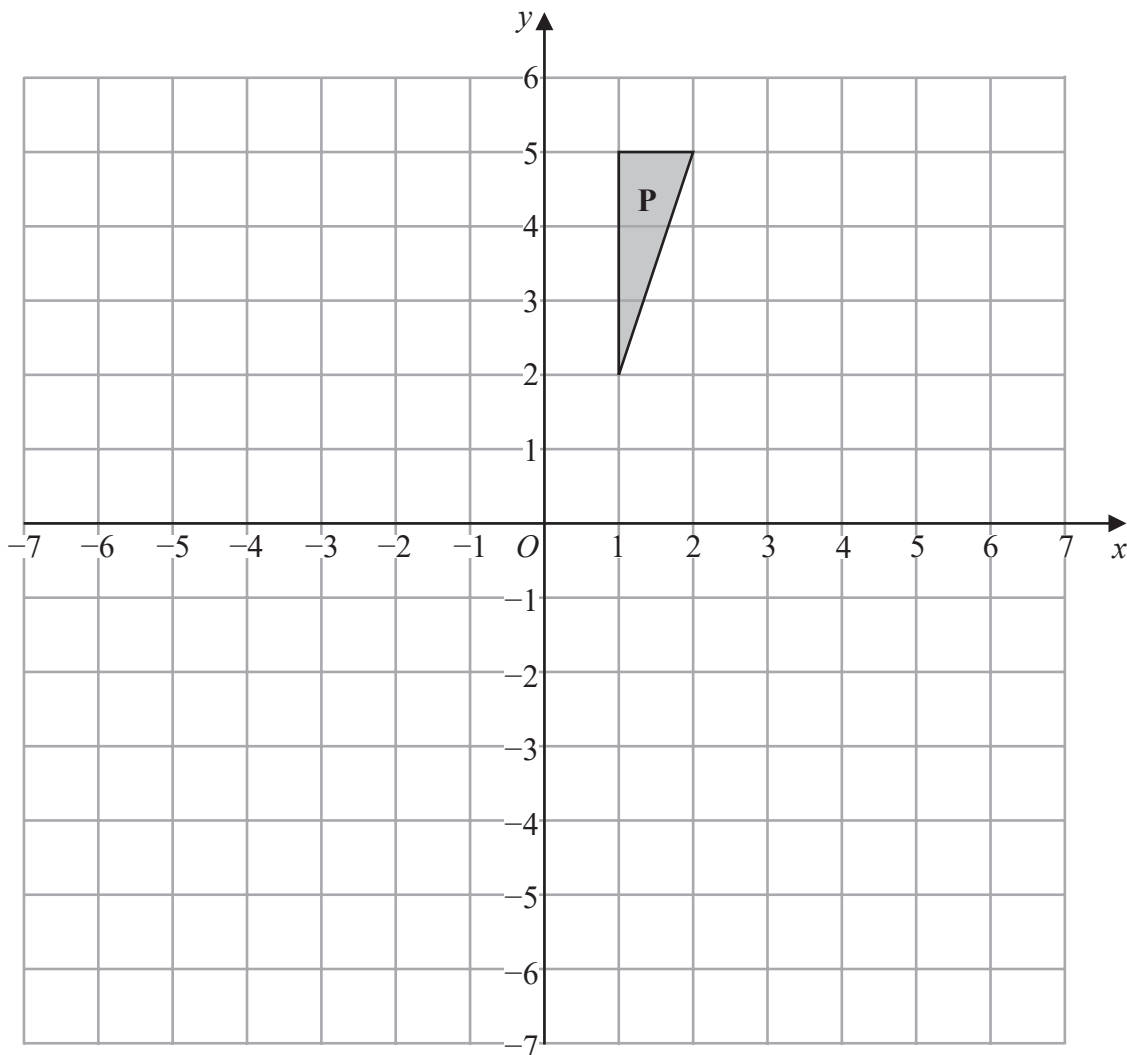


11 The diagram shows a triangle **P** on a grid.



Triangle **P** is rotated 180° about $(0, 0)$ to give triangle **Q**.

Triangle **Q** is translated by $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ to give triangle **R**.

(a) Describe fully the single transformation that maps triangle **P** onto triangle **R**.

(3)

Under the transformation that maps triangle **P** onto triangle **R**, the point *A* is invariant.

(b) Write down the coordinates of point *A*.

(,)

(1)

(Total for Question 11 is 4 marks)